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GRADUATION PROJECT

Design and Development of an Autonomous Restaurant Service Robot for Order Delivery and Customer Interaction

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List of Abbreviations and Terms

Abbreviation	Full term	Definition
A*	A-star	Graph-search algorithm that finds a minimum-cost path on a map.
DWA	Dynamic Window Approach	Local planner that chooses safe, reachable velocity commands.
EKF	Extended Kalman Filter	Estimator that fuses noisy sensor data for nonlinear systems.
E-Stop	Emergency Stop	Switch that immediately halts the robot motion.
FOPI	Fractional-Order PI controller	PI controller whose integral action has a non-integer order.
GA	Genetic Algorithm	Optimisation method inspired by natural selection.
HSR	Human Support Robot	Mobile manipulator produced by Toyota for service research.
IMU	Inertial Measurement Unit	Sensor that measures angular rate and acceleration.
IR	Infrared	Radiation used by simple proximity and line sensors.
ITAE	Integral of Time-weighted Absolute Error	Index used to tune controllers.
LiDAR	Light Detection and Ranging	Sensor that measures distances with laser light.
LLM	Large Language Model	Neural network that understands and generates language.
QR	Quick Response code	Two-dimensional barcode that stores a link or short data.
ROS	Robot Operating System	Open-source middleware for building robot software.
SLAM	Simultaneous Localisation and Mapping	Building a map while estimating the robot position in it.
SMART	Specific, Measurable, Achievable, Relevant, Time-bound	Criteria used to formulate project objectives.

Chapter 1

General Introduction

This chapter introduces the project, states the problem it addresses, and presents the objectives and the technical specifications that were defined on the basis of the literature review in Chapter 2.

1-1 Project Overview and Background

A service robot is a robot that performs useful tasks for humans or equipment outside industrial automation [1]. Restaurants are a natural field for such robots, because waiting staff spend much of every shift carrying food between the kitchen and the tables. This project aims to design and develop an autonomous wheeled robot that delivers orders from the kitchen to the tables, is linked to the restaurant ordering workflow, and interacts clearly with customers.

1-2 Problem Statement and its Importance

- **Practical problem:** carrying food from the kitchen to the tables is slow, repetitive and tiring, especially at peak times, and restaurants increasingly lack staff.
- **Technical problem:** the reviewed studies solve only part of the task. Low-cost robots follow lines on the floor and cannot adapt to layout changes [2], map-based robots lack an ordering system [3], and the most flexible robot relies on an expensive platform [4].

Solving this problem helps restaurants keep service running with fewer staff, and it combines several core topics of robotics and intelligent systems in one application.

1-3 Project Objectives (Main and Secondary)

The objectives were derived from the literature review and written following the SMART approach. The core objectives are to be achieved in the first complete prototype, and the secondary objectives before the final project submission.

1-3-1 Main Objective

To design and build a low-cost autonomous restaurant service robot that navigates between tables without floor markings, integrates with the restaurant workflow from order to pickup confirmation, interacts clearly with customers, and is evaluated with measurable results.

1-3-2 Core Objectives

A. Navigation and motion

- **O1 – Map-based autonomous navigation:** reach any mapped table without floor lines, with at least 95% success over 50 trials and a stopping error within ±10 cm. This removes the main limitation of line-following robots [2].
- **O2 – Safe obstacle avoidance:** avoid static and moving obstacles with zero collisions over 50 trials among moving people, keeping at least 30 cm from any person.
- **O3 – Accurate localisation:** fuse wheel, IMU and LiDAR data so that the position error at the table is below 10 cm and the heading error below 5°, following the fusion approach of [3].
- **O4 – Smooth motor control:** each wheel speed loop settles in less than 0.3 s with overshoot below 5%, so that drinks do not spill.

B. Service system

- **O5 – Integrated digital ordering:** a QR-based ordering system delivers at least 99% of orders to a kitchen dashboard in less than 3 s and forwards a delivery command to the robot, as in [2].
- **O7 – Pickup confirmation and automatic return:** detect that the customer has taken the food with at least 97% accuracy before the robot returns or continues.

C. Interaction

- **O10 – Clear status interface:** a screen and voice messages show what the robot is doing, with a clarity score of at least 4 out of 5 from test users, as recommended in [4].

D. Hardware and safety

- **O11 – Low-cost hardware:** build the robot from locally available components within a budget fixed in advance, with a full bill of materials.
- **O12 – Sturdy body, payload and battery:** carry at least 3 kg and work at least 3 h continuously at full load.
- **O13 – Safety:** a physical and software emergency stop that halts the robot in less than 0.5 s, and a maximum speed of 0.5 m/s.

E. Testing and documentation

- **O14 – Quantitative test plan:** test each core function at least 50 times under documented conditions and report mean ± standard deviation.
- **O15 – Real-user evaluation:** test the robot with at least 30 participants using a five-point questionnaire and a statistical test.
- **O16 – Consistent documentation:** no contradiction between text and tables, and limitations stated openly.

1-3-3 Secondary Objectives

- **O6 – Multi-order delivery:** deliver at least 3 orders to different tables in one trip, which addresses the one-order limit of [2].
- **O8 – LLM task planning on low-cost hardware:** a large language model selects the next robot skill from a fixed list with at least 90% correct choices and a response time below 3 s, with an offline fallback mode, following [4].
- **O9 – One-button error correction:** a "something went wrong" button triggers replanning, with at least 70% of reported errors corrected.

1-4 Project Scope and Technical Specifications

Scope. The project covers a wheeled mobile base with shelves, its sensors and controllers, navigation software, a web-based ordering system linked to the kitchen and the robot, and a status interface; the LLM-based planner is a secondary objective. It does not cover food preparation, a robotic arm, payment-gateway integration, several robots, outdoor operation or moving between floors.

Technical specifications. Table 1-1 lists the minimum technical specifications that the robot must satisfy, following the faculty template.

Table 1-1: Technical specifications of the proposed robot

Category	Parameter	Specification
Operating environment	Workspace	Indoor restaurant, flat and smooth floor, aisles ≥ 70 cm
	Conditions	Fixed or variable indoor lighting; static obstacles and moving people
Physical	Dimensions	Footprint ≤ 50 × 50 cm, height ≤ 120 cm; passes a 60 cm opening
	Mass	Total mass ≤ 25 kg
	Workspace	Any mapped table; tray height 75–110 cm
Payload and dynamics	Payload	Carry ≥ 3 kg on 3 shelves
	Dynamics	Acceleration ≤ 0.5 m/s² so that drinks do not spill
Actuators and power	Speed	Linear ≥ 0.3 m/s (capped at 0.5 m/s); angular ≥ 0.8 rad/s at full load
	Positioning accuracy	Stop at the table within ±100 mm and ±5°
	Power autonomy	≥ 3 h of continuous work at full load
Sensors	Obstacle detection	360° LiDAR, range 0.15 m to 6 m
	Orientation	IMU heading accuracy ±1°
Processing	Update rate	Navigation and obstacle avoidance ≥ 10 Hz
	Latency	Perception to action ≤ 100 ms
Communication	Data and range	Wi-Fi; sensor and status data ≥ 10 msg/s; stable over ≥ 30 m
	Interface	Touchscreen with status, battery and errors; manual/automatic switch
Safety and reliability	E-Stop	Physical and software E-Stop; stop in < 0.5 s
	Protection	Fuses against over-current; insulated, shielded wiring against EMI
	Safe mode	Stop on Wi-Fi loss or obstacle < 30 cm; return to base at battery < 15%

1-5 Report Structure

This report contains two chapters. Chapter 1 introduces the project, its objectives and its technical specifications. Chapter 2 reviews three recent studies on restaurant service robots, compares them and identifies the research gap.

1-6 Conclusion

This chapter presented the problem addressed by the project, its main objective, thirteen core and three secondary SMART objectives, and the minimum technical specifications. The next chapter reviews the studies on which these objectives are based.

Chapter 2

Literature Review and Previous Studies

This chapter summarises three recent studies on restaurant service robots, compares them in terms of features, technologies, strengths and weaknesses, and identifies the research gap that the project addresses.

2-1 Previous Studies in the Field of the Project

The three studies were selected because they target restaurant robots specifically, were published between 2024 and 2026, were implemented on real prototypes, and each address a different layer of the problem: motion, service and interaction.

2-1-1 Map-Based Autonomous Navigation (Mac et al., 2025)

Mac et al. built a low-cost restaurant robot on a Raspberry Pi 4 and an Arduino running ROS [3]. It uses RTAB-Map visual SLAM with an RGB-D camera fused with an IMU through an EKF, A* for global planning, DWA for local obstacle avoidance and YOLOv3 for obstacle detection, while a GA-tuned fractional-order PI controller drives the wheels. Settling time fell from 0.58 s to 0.26 s and rise time from 0.35 s to 0.17 s, with each test repeated ten times.

Limitations: off-the-shelf components with default settings, no obstacle-detection accuracy, a small test area, an abstract that contradicts the reported overshoot, and no ordering system or customer interaction.

2-1-2 Complete Restaurant Service Cycle (Sayor et al., 2024)

Sayor et al. presented RoboServe [2]. Customers scan a table QR code, order and pay online, and the order reaches a chef dashboard; an ESP32 robot then follows a black floor line with IR sensors to the table, and an IR sensor detects the empty tray to send it back. Over 100 attempts per feature, it reached 97% correct-table delivery and 99% order taking at a cost of about 5,600 BDT.

Limitations: line-following navigation, a timed obstacle detour without distance sensing, one order per trip with a 1.5 kg payload, and success rates without test conditions.

2-1-3 LLM Task Planning with One-Button Recovery (Terashima et al., 2026)

Terashima et al. built on earlier work by the same group [5] and proposed a waiter robot on a Toyota HSR in which GPT-4o chooses the next skill step by step, and a single tablet button lets the user report a failure so the robot replans [4]. With 224 users in a simulated restaurant, task success rose from 55% to 86% and customer anxiety fell from 3.1 to 2.2 out of 5.

Limitations: a preprint not yet peer-reviewed, different user groups without a statistical test, no description of navigation, and a very expensive platform.

2-2 Comparative Analysis of Existing Systems

Table 2-1 compares the three systems in terms of features and technologies, and Table 2-2 summarises their main strengths and weaknesses.

Table 2-1: Comparison of the studied systems

Criterion	Mac et al. (2025)	RoboServe (2024)	Terashima et al. (2026)
Publication type	Peer-reviewed journal (Elsevier)	Peer-reviewed conference (ACM)	Preprint (arXiv)
Platform	Raspberry Pi 4, Arduino, ROS	ESP32	Toyota HSR, GPT-4o
Cost	Low	Very low	Very high
Navigation	Advanced: SLAM, A*, DWA	Primitive: line following	Not described
Ordering and payment	None	Complete: QR, payment, chef dashboard	Voice ordering, no payment
Intelligence	Weak	Moderate: voice and touch	Advanced: LLM planning and recovery
Error handling	None	Voice alert at obstacles	Replanning after button press
User testing	None	None	224 users and questionnaire

Criterion	Mac et al. (2025)	RoboServe (2024)	Terashima et al. (2026)
Evaluation rigour	Good: repeated tests, mean ± SD	Weak: no test conditions	Moderate: no statistical test
Headline result	Settling time 0.58 → 0.26 s	Correct delivery 97%	Task success 55% → 86%

Table 2-2: Main strengths and weaknesses of the studied systems

Study	Main strengths	Main weaknesses
Mac et al. (2025)	• Integrated ROS navigation (SLAM, EKF, A*, DWA) • GA-tuned controller with a clear response gain • No floor lines needed	• No algorithmic novelty • No obstacle-detection accuracy • No ordering system or interaction
RoboServe (2024)	• Full cycle from order to payment • Automatic return by empty-tray detection • Very low cost and simple algorithms	• Inflexible line following • Unsafe timed obstacle detour • One order per trip, 1.5 kg payload
Terashima et al. (2026)	• LLM task planning (GPT-4o) • One-button error recovery • Tested with 224 users	• Not peer-reviewed • Simulated restaurant, no statistics • Expensive robot

2-3 Research Gap

Figure 2-1 summarises how the three studies cover the three layers of the problem. Each is strong in only one layer, and none combines autonomous navigation, a complete service system and intelligent error recovery in one low-cost robot. This gap is the basis of the objectives in Chapter 1.

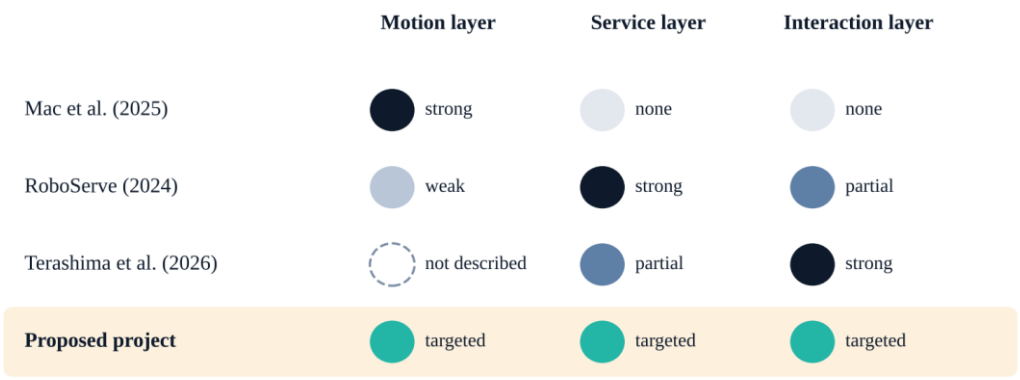


Figure 2-1: Coverage of the motion, service and interaction layers by the studied systems and the proposed project

2-4 Conclusion

This chapter reviewed three recent restaurant robot studies, compared them and showed that none of them combines motion, service and interaction on low-cost hardware, which defines the research gap of the project.

References

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